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foundation

Carbon Emission Prediction

Learning Objectives

- Understand the causes and impact of carbon emissions
- Explore and clean real-world carbon emission datasets
- Apply machine learning models to predict emission levels
- Analyze key factors influencing emission trends
- Improve data handling and model evaluation skills



Tools and Technology

- **Python** – For data preprocessing, analysis, and machine learning model building
- **Pandas & NumPy** – To handle and manipulate large datasets efficiently
- **Scikit-learn** – For training and evaluating ML models like Linear Regression, Random Forest
- **Matplotlib & Seaborn** – To visualize emission trends and model performance
- **Jupyter Notebook / Google Colab** – As the primary development environment for experimentation

Methodology

- **Data Collection:** Acquired emission data from public sources
- **Preprocessing:** Cleaned and prepared the data for analysis
- **EDA:** Explored patterns and visualized key insights
- **Model Training:** Applied ML algorithms for prediction
- **Evaluation:** Tested model performance using accuracy metrics

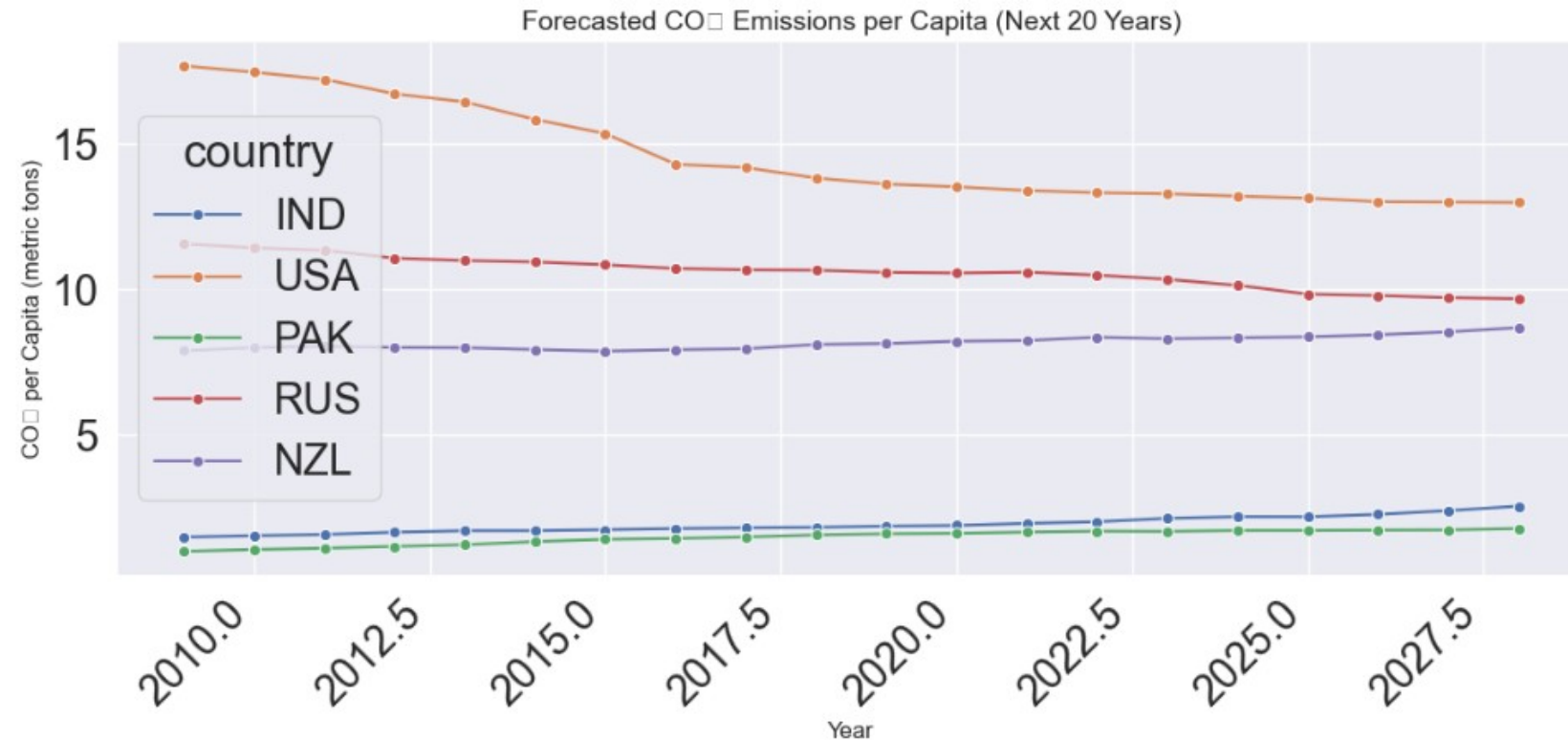
Problem Statement:

Analysis of country-specific data and development of machine learning models in order to predict CO2 emissions from country parameters. The project uses the publicly available dataset Climate Change Data from the World Bank Group.

Solution:

- Cleaned and processed climate and economic data from the World Bank (1990–2011).
- Performed data analysis to identify key factors influencing CO₂ emissions.
- Built and trained a Random Forest Regressor model to accurately predict emissions.

Screenshot of Output:



Forecasted CO₂ per Capita for Last 5 Years in Forecast Period (India):

	country	year	co2_percap
15	IND	2024	2.194019
16	IND	2025	2.190429
17	IND	2026	2.272474
18	IND	2027	2.401276
19	IND	2028	2.555614

Conclusion:

- CO₂ emissions per capita are projected to decrease in developed countries like the USA and Russia, reflecting efforts in sustainability, clean energy adoption, and stricter emission regulations.
- India and Pakistan show a rising trend, indicating the environmental impact of rapid industrialization and economic growth.
- The forecast highlights a convergence in global emission patterns, though countries are at different stages of development.
- These insights emphasize the need for balanced growth, where emerging economies like India adopt greener technologies while sustaining development.